Work Sheet 7

Practice Problems

1 Center of Circle : $(2,-2,1) = c$

Point : $(1,2,4) = p$

$\vec{cp} = \langle -1, 4, 3 \rangle$

distance : $\sqrt{(-1)^2 + 4^2 + 3^2} = \sqrt{26}$

2 $|\vec{a}| = 12 \quad |\vec{b}| = 3 \quad \vec{a} \cdot \vec{b}$ (max) $= 36$

$\vec{a} = \langle \pm 12, 0 \rangle \quad \vec{b} = \langle \pm 3, 0 \rangle$

3 $\vec{a} = \langle 7, -3, 4 \rangle \quad \vec{b} = \langle -2, 6, 8 \rangle$

$\vec{a} \times \vec{b} = \langle -48, -64, 36 \rangle = \vec{i}$

$|\vec{a} \cdot \vec{i}| = (7)(-48) + (-3)(-64) + (4)(36) = 0$

$|\vec{b} \cdot \vec{i}| = (-2)(-48) + (6)(-64) + (8)(36) = 0$

$\vec{b} \times \vec{a} = \langle 48, 64, -36 \rangle = \vec{j}$

$$\begin{matrix} 7 & -3 & 4 & 7 \\ -2 & 6 & 8 & -2 \\ 7 & -3 & 4 & 7 \end{matrix}$$

Challenge Problems

8 $3x^2 + 3y^2 + 3z^2 = 10 + 6y + 12z$

$3x^2 + 3y^2 + 6y + \underline{\quad} + 3z^2 + 12z + \underline{\quad} = 10$

$3x^2 + 3(y^2 + 2y + \underline{1}) + 3(z^2 + 4z + \underline{4}) = 10 + \underline{1} + \underline{4}$

$3x^2 + 3(y+1)^2 + 3(z+2)^2 = 15$

Center : $(0, -1, -2)$ radius $= \sqrt{15}$

11 $a = (2,4,2) \quad b = (3,7,2) \quad c = (1,3,5)$

$\vec{ab} = \langle -1, 3, 0 \rangle \quad \vec{bc} = \langle -2, -4, 3 \rangle$

if on same line then $\parallel$

if $\parallel$ then $\vec{ab} \cdot k = \vec{bc}$

k is some constant that when $\vec{ab}$ is multiplied by k the resulting vector is $\vec{bc}$. There is no possible constant k so the points a, b, c are not colinear